PhyEdit: Towards Real-World Object Manipulation via Physically-Grounded Image Editing

Ruihang Xu
ruihangxu@zju.edu.cn
ReLER, CCAI, Zhejiang University
Hangzhou, China

Dewei Zhou
zdw1999@zju.edu.cn
ReLER, CCAI, Zhejiang University
Hangzhou, China

Xiaolong Shen
sxlongcs@zju.edu.cn
ReLER, CCAI, Zhejiang University
Hangzhou, China

Fan Ma
flowerfan524@gmail.com
ReLER, CCAI, Zhejiang University
Hangzhou, China

Yi Yang
yangyics@zju.edu.cn
ReLER, CCAI, Zhejiang University
Hangzhou, China

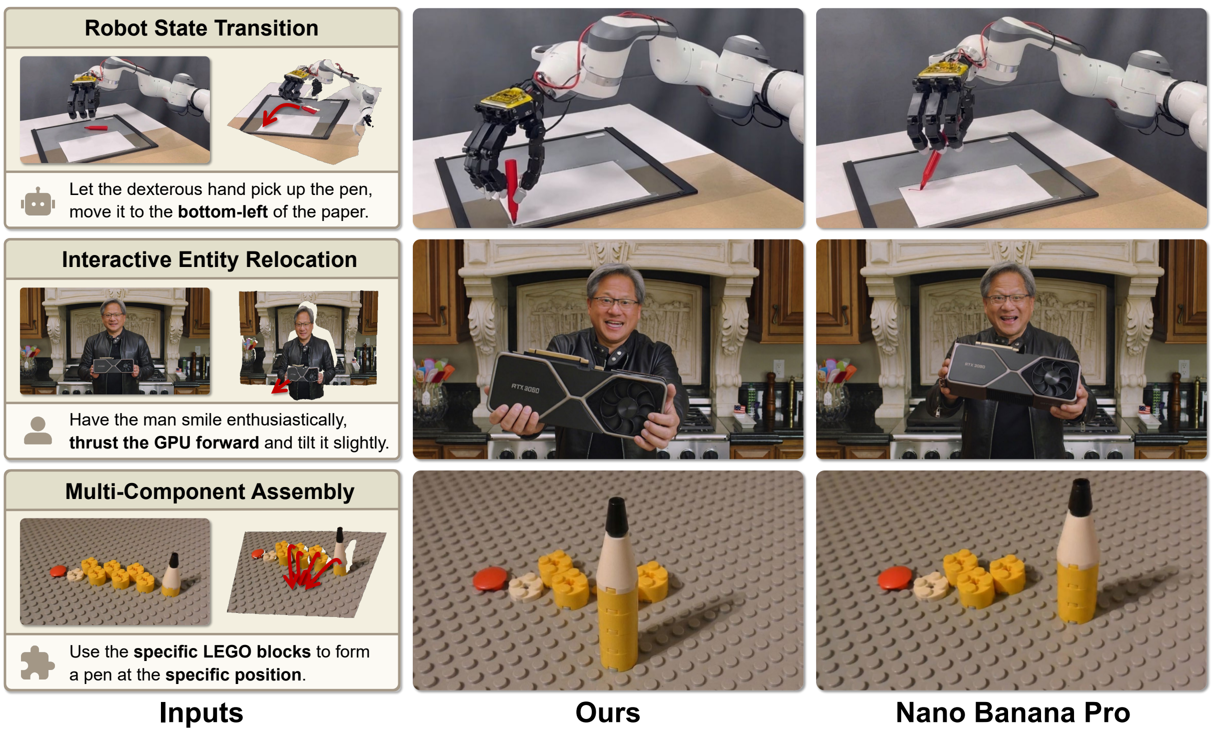

Figure 1: Comparison of image editing results on three physical scenarios between our PhyEdit and Nano Banana Pro [11]

Abstract

Achieving physically accurate object manipulation in image editing is essential for its potential applications in interactive world models. However, existing visual generative models often fail at precise spatial manipulation, resulting in incorrect scaling and positioning of objects. This limitation primarily stems from the lack of explicit mechanisms to incorporate 3D geometry and perspective projection. To achieve accurate manipulation, we develop PhyEdit, an image editing framework that leverages explicit geometric simulation as contextual 3D-aware visual guidance. By combining this plug-and-play 3D prior with joint 2D–3D supervision, our method effectively improves physical accuracy and manipulation consistency. To support this method and evaluate performance, we present a real-world dataset, RealManip-10K, for 3D-aware object manipulation featuring paired images and depth annotations. We also propose ManipEval, a benchmark with multi-dimensional metrics to evaluate 3D spatial control and geometric consistency. Extensive experiments show that our approach outperforms existing methods, including strong closed-source models, in both 3D geometric accuracy and manipulation consistency.

CCS Concepts

• **Computing methodologies → Computer vision tasks**; *Image manipulation.*

Keywords
image editing, physically-grounded editing, object manipulation

1 Introduction

Diffusion models have achieved impressive results in image generation and editing. Recent visual generative models [6, 11, 25, 33, 39] perform seamless editing such as manipulating an object at the 2D level. These general models and those specialized for object manipulation [21, 22, 43, 57, 59] execute precise spatial translation when the prompt contains detailed coordinate instructions (e.g., 'move the object to position (x, y)'). However, 2D-level manipulation is often insufficient for emerging real-world applications. For instance, object-centric robotic manipulation guided by visual priors [3, 36, 62] requires precise physical manipulation in 3D space. This demands models to act as interactive world models capable of rendering 3D-aware and geometrically consistent state transitions. Achieving such precise rendering remains a significant challenge.

The limitations of existing models in acting as reliable state renderers stem from two main aspects. **(1) Lack of Explicit 3D and Physical Control.** Recent large visual generative models attempt to learn world dynamics from visual data [5, 20, 24]. However, without explicit mechanisms to incorporate perspective projection laws, these models often produce physically implausible behaviors, such as incorrect scale variations during object movement and inconsistent trajectories. In practical applications, this limitation prevents precise object manipulation, restricting users to vague textual prompts like "further" or "closer", or coarse scaling factors, making fine-grained geometric manipulation unattainable. **(2) Lack of High-Quality Real-World Datasets and Benchmarks.** Existing datasets for image editing [7, 43, 59] place little emphasis on 3D-aware spatial changes and real-world physical laws. Meanwhile, 3D asset datasets [12, 30, 57] are primarily synthesized using 3D engines. They contain few non-rigid real-world objects. This limits the generalization of trained models in simulating the real world. Furthermore, related benchmarks [22, 30, 44, 59] typically evaluate 2D metrics and overlook depth accuracy. To facilitate research in 3D image editing, both a high-quality real-world dataset and a benchmark with appropriate 3D evaluation metrics are required.

To bridge this gap, we introduce **PhyEdit**, taking a step towards real-world object manipulation by framing 3D-aware editing as generative state rendering. Driven by explicit 3D movement instructions from either users or interactive systems, our DiT-based approach handles geometric displacement while naturally accounting for implicit physical effects such as deformations and occlusions. Specifically, our framework features a plug-and-play contextual 3D-aware visual guidance module to inject geometric priors, and a non-intrusive joint loss strategy utilizing both 2D and 3D supervision to improve spatial accuracy.

Along with the framework, we propose **RealManip-10K**, a high-quality real-world dataset tailored for 3D-aware object manipulation. This dataset provides real-world image pairs demonstrating physical object manipulation in 3D space, along with detailed depth annotations. We also design a specialized benchmark, **ManipEval**, with representative metrics to evaluate spatial control accuracy.

Our method achieves state-of-the-art performance on ManipEval, outperforming existing baselines in 3D manipulation precision. Ablation studies further validate the effectiveness of each proposed module. The dataset, benchmark, and training code will be made publicly available upon publication.

In summary, our key contributions are threefold:

- **Method**: We develop **PhyEdit**, a DiT-based framework designed for physically-accurate image editing. It utilizes a contextual 3D visual guidance module and 2D-3D joint supervision to improve manipulation accuracy.
- **Dataset and Benchmark**: We construct **RealManip-10K**, a high-quality real-world dataset for 3D-aware object manipulation, and **ManipEval**, a dedicated benchmark with metrics for precisely measuring 3D spatial accuracy.
- **SOTA Performance**: Our approach achieves superior performance on ManipEval compared to existing baselines. Ablation studies confirm the effectiveness of our design choices.

2 Related Work

2.1 Object Manipulation

In the context of interactive world models, manipulating the spatial arrangement of objects within an image can be viewed as rendering a state transition.

2D Object Manipulation. To achieve such transitions, existing approaches operating in the 2D plane generally fall into two categories: drag-based methods and explicit manipulation. Drag-based methods [27, 34, 44, 59] allow users to shift visual content by pulling specified control points, but they typically require time-consuming per-image optimization during inference. Alternatively, explicit manipulation pipelines [13, 21, 22] process the target subject as a coherent entity through extraction, translation, and background inpainting, yet they often struggle with complex occlusions and multi-object interactions. Although both paradigms effectively shift objects across the image plane, they operate inherently in 2D and lack a structural understanding of 3D scene geometry.

3D Object Manipulation. To address this limitation, another line of research tackles manipulation from a 3D perspective. Early explicit methods [10, 47, 60] bypass 2D limitations by requiring full 3D scene reconstruction and subsequent re-rendering. However, their final image quality depends heavily on the accuracy of the reconstructed geometry and lighting, and they often fail with non-rigid deformations. More recent works avoid heavy reconstruction by incorporating 3D spatial information or visual generative priors directly into the editing process. For instance, some methods apply 3D transformations within the latent space [35, 42], encode 3D assets and positional data into tokens [55], or use video generation to guide spatial movements [54, 57]. Our work follows this generative line of research but emphasizes physically-grounded simulation to achieve precise 3D object manipulation.

2.2 Datasets and Benchmarks for Object Manipulation

Current real-world image editing datasets [7, 9, 49] mostly focus on motion or shape changes. They lack samples that clearly reflect physical laws, such as perspective-induced size variations caused by depth movements. On the other hand, 3D simulation datasets [1, 30] offer limited visual diversity and typically focus on a single, isolated

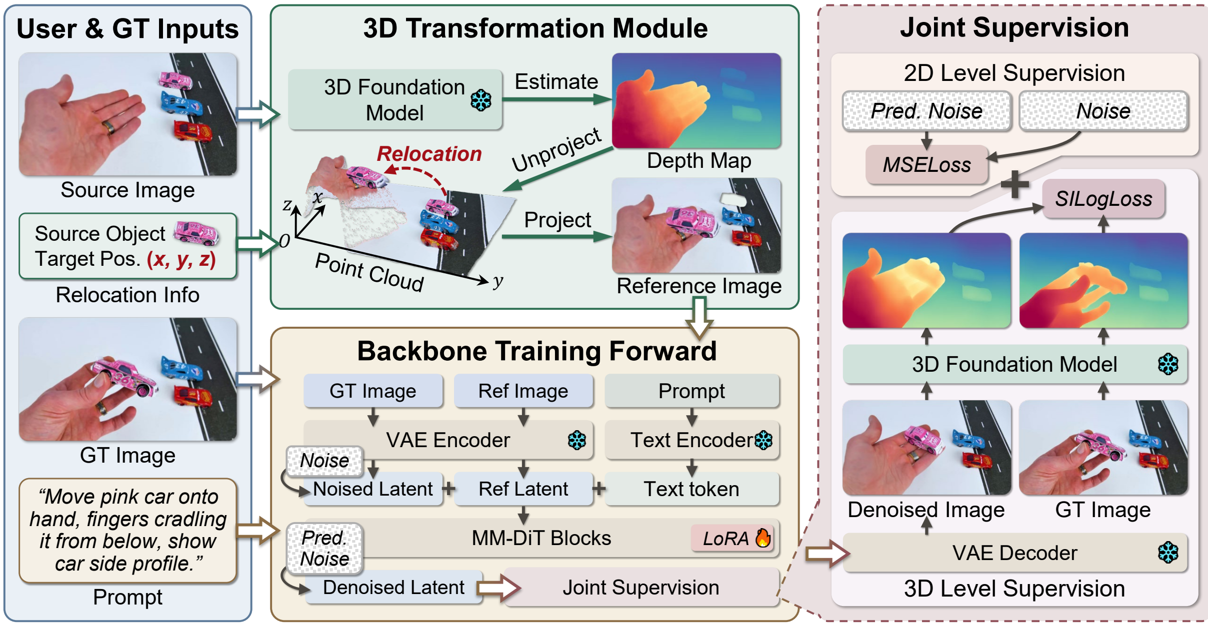

Figure 2: Overview of PhyEdit. User and GT inputs are first processed via the 3D transformation module. The resulting conditions are then fed into the backbone for the training forward pass, followed by 2D and 3D level joint supervision.

object. They cannot reflect the complex interactions among multiple objects during real-world state transitions. For evaluation, existing benchmarks [22, 30, 34, 44] mostly measure image feature distances and general quality, rather than testing control accuracy in actual physical and geometric dimensions.

3 Method

3.1 Preliminaries

Diffusion Transformer for Image Editing. Diffusion Transformer (DiT) models [37] are widely used in image generation and editing because they support flexible multi-modal conditioning. For one denoising step, the DiT input token sequence is

$$\mathbf{T} = \left[\mathbf{T}_{\text{text}}; \mathbf{T}_{\text{img}}; \mathbf{T}_{\text{cond}}\right], \tag{1}$$

where $\mathbf{T}_{\text{text}}$ is the text-prompt embedding, $\mathbf{T}_{\text{img}}$ is the noisy image latent token, and $\mathbf{T}_{\text{cond}}$ denotes additional condition tokens. Here $[\cdot;\cdot]$ means concatenation.

Given clean latent $\mathbf{z}_0$ and noise $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$, flow matching defines a linear interpolation path parameterized by $t \in [0, 1]$:

$$\mathbf{z}_t = (1-t)\mathbf{z}_0 + t\boldsymbol{\epsilon}. \tag{2}$$

The corresponding ground-truth velocity along this path is $\mathbf{v}^* = \boldsymbol{\epsilon} - \mathbf{z}_0$. The DiT is trained to regress this velocity:

$$\mathcal{L}_{\text{flow}} = \mathbb{E}_{t,\mathbf{z}_0,\boldsymbol{\epsilon}}\left[\|v_\theta(\mathbf{T}, t) - (\boldsymbol{\epsilon} - \mathbf{z}_0)\|_2^2\right]. \tag{3}$$

Here v_θ is the DiT velocity predictor, and the loss matches predicted velocity to the ground-truth transport direction.

Multi-View 3D Foundation Model. A multi-view 3D foundation model jointly predicts scene geometry and camera parameters from a set of input images. Recent methods such as VGGT [50], Pi3 [51], and Depth-Anything-3 [26] adopt a Transformer followed by some prediction heads [41]. Given N input images $\{I_i\}_{i=1}^N$, each image is encoded and paired with a learnable camera token; the full sequence is processed jointly:

$$\left[\{z_{\text{cam},i}\}; \{\mathbf{Z}_i\}\right] = F\left(\left[\{x_{\text{cam},i}\}; \{E(I_i)\}\right]\right), \quad i = 1, \dots, N. \tag{4}$$

$E(\cdot)$ is the shared image encoder, $x_{\text{cam},i}$ is the learnable camera token for view i, and F is the cross-view transformer. It outputs per-view camera features $z_{\text{cam},i}$ and visual tokens $\mathbf{Z}_i$. Prediction heads output depth and camera pose for each view:

$$D_i = H_d(\mathbf{Z}_i), \qquad (R_i, t_i) = H_{\text{cam}}(z_{\text{cam},i}), \tag{5}$$

where H_d and H_{cam} are shared prediction heads, D_i is the depth map for view i, and (R_i, t_i) is its camera rotation and translation.

3.2 Model Architecture

Overall Architecture. To use 3D priors effectively for object manipulation, we combine a DiT editing backbone with a 3D foundation model (Fig. 2). The framework has three parts: (1) a 3D transformation module that generates a depth-aware preview, (2) the DiT denoising backbone, and (3) a joint training loss in 2D latent and 3D depth spaces.

3D Transformation Module. Given source image I_{src}, object mask M_o (manual or automatic), and transition vector $\Delta\mathbf{p}_o$, we first predict depth D and camera pose (R, t), then edit the object directly in 3D space:

$$\mathbf{P}_o = \text{Unproj}(I_{\text{src}}, M_o, D, R, t), \tag{6}$$

$$\mathbf{P}'_o = \mathbf{P}_o + \Delta\mathbf{p}_o, \tag{7}$$

$$I_{\text{prev}} = \text{Proj}(\mathbf{P}'_o, R, t). \tag{8}$$

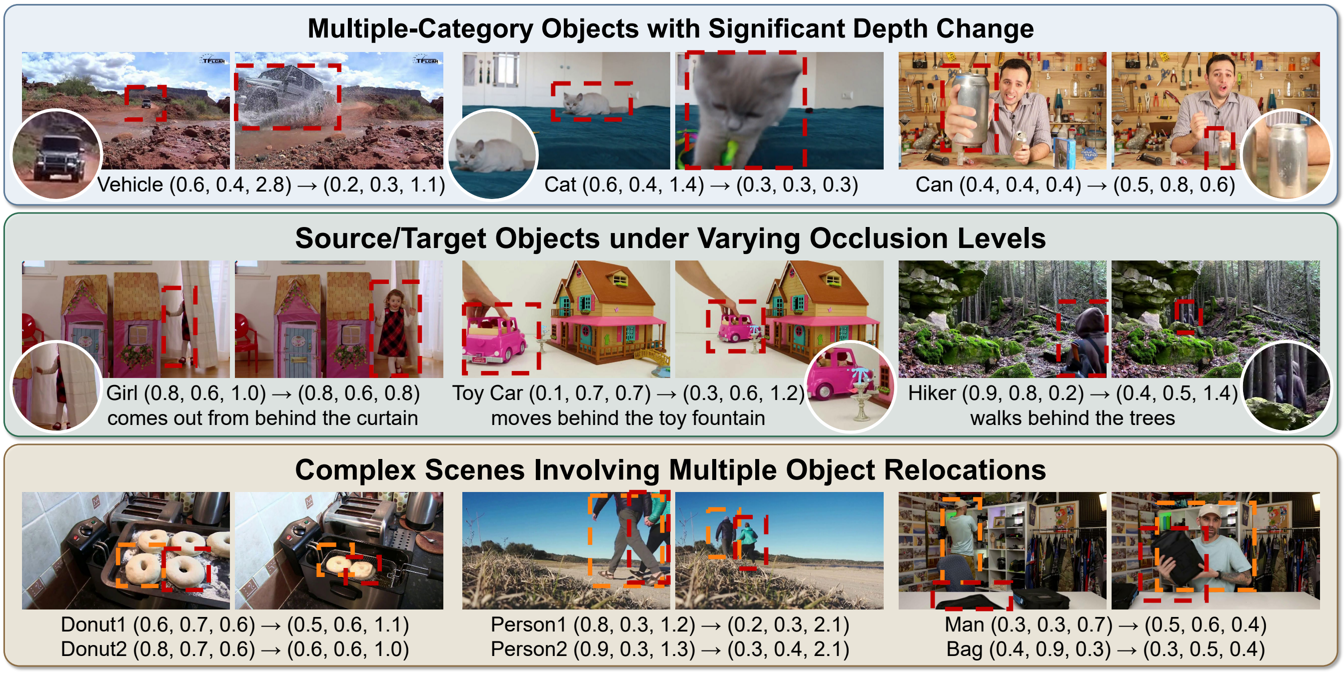

Figure 3: Samples from RealManip-10K. The objects are annotated with red and orange boxes. And the 3D object coordinates (x, y, z) are under the camera coordinate system, where the origin is at the top-left-near corner.

The preview image I_{prev} is used as an additional condition for DiT. This gives explicit geometric guidance and naturally supports multi-object manipulation without iterative editing.

Backbone Training Forward. During training, the DiT input combines text, noisy latent, and preview condition:

$$\mathbf{T} = \left[\mathbf{T}_{\text{text}}; \mathbf{T}_{\text{img}}; \mathbf{T}_{\text{prev}}\right], \tag{9}$$

$$v_\theta(\mathbf{T}, t) = \text{MM-DiT}(\mathbf{T}, t). \tag{10}$$

Joint Supervision. Latent-space denoising loss alone is often insufficient for 3D manipulation, because it emphasizes appearance reconstruction more than geometric correctness. We therefore add a depth-space supervision term.

We first estimate the clean latent and decode the edited image:

$$\hat{\mathbf{z}}_0 = \mathbf{z}_t - v_\theta(\mathbf{T}, t), \tag{11}$$

$$\hat{I}_{\text{edit}} = \text{Dec}(\hat{\mathbf{z}}_0), \tag{12}$$

$$\mathcal{L}_{\text{depth}} = \text{SILog}\left(D_{\text{edit}}, D_{\text{gt}}\right), \tag{13}$$

where D_{edit} and D_{gt} are depth maps of the edited and target images. We then use the scale-invariant logarithmic loss [14]:

$$\begin{aligned} \text{SILog}(D_{\text{pred}}, D_{\text{gt}}) &= \frac{1}{N}\sum_{i=1}^{N}\Delta_i^2 - \frac{1}{N^2}\left(\sum_{i=1}^{N}\Delta_i\right)^2, \\ \Delta_i &= \log D_{\text{pred}}(i) - \log D_{\text{gt}}(i), \end{aligned} \tag{14}$$

where N is the number of valid depth pixels. The final objective is

$$\mathcal{L} = \mathcal{L}_{\text{noise}} + \lambda_d \mathcal{L}_{\text{depth}}, \tag{15}$$

where λ_d balances latent and depth supervision.

Compared with methods that require heavy architecture changes to inject 3D priors [42, 55, 58], our approach only adds lightweight conditioning and training design. It can be plugged into different DiT-based editors with minimal modification.

3.3 Dataset Construction

Dataset Overview. As discussed in Sec. 2.2, existing datasets are not sufficient for 3D-aware object manipulation. We therefore build RealManip-10K, a real-world dataset of paired images $(I_{\text{src}}, I_{\text{tgt}})$ where one or more objects are manipulated in 3D space, especially along the depth axis. Each pair provides depth maps, object masks, and representative 3D object coordinates.

As shown in Fig. 3, the dataset covers diverse scenes and object types. Several features make it suitable for 3D manipulation learning: (1) Each pair has an object with significant depth change. (2) The dataset includes samples with part of the object occluded, which emphasizes the near-far relationship within the scene. (3) The dataset contains pairs with multiple objects manipulated simultaneously, which encourages the model to learn complex spatial interactions. The construction pipeline is shown in Fig. 4. We detail the pipeline in the following parts.

Data Source. Data quality is critical: low-quality sources can weaken the supervision signal and hurt generation quality. We collect videos from high-resolution and minimally overlapping datasets, including *OpenVid-1M* [32], *VIDGEN-1M* [46], and *PE-Video-Dataset* [4]. We also include object-tracking datasets (*LaSOT* [16], *GoT-10K* [19] and *TrackingNet* [31]) as supplements. All clips are filtered by video quality assessment (VQA) [53] and motion estimation [52].

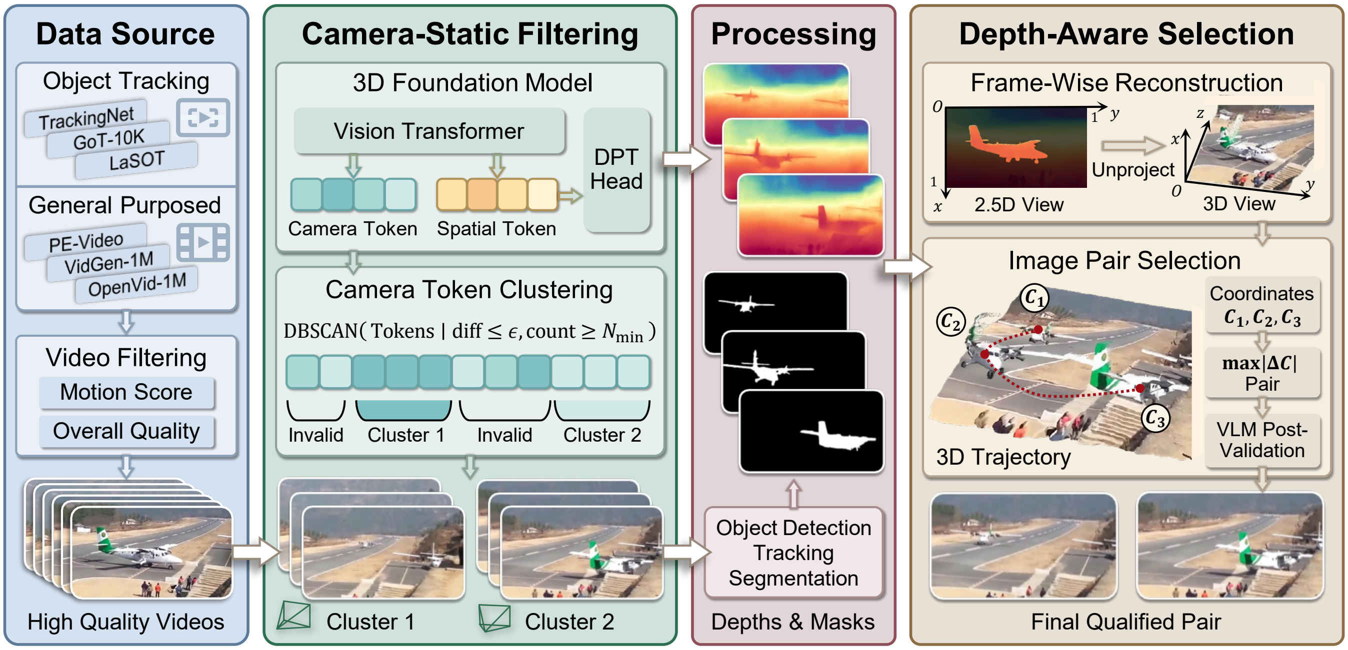

Figure 4: Dataset Construction Pipeline. The workflow consists of four key stages: data source filtering, camera-static clip extraction, depth and mask processing, and depth-aware frame pair selection. At the bottom of the second column, the symbols and represent 3D viewing frustums, denoting that Cluster 1 and Cluster 2 have different camera viewpoints.

Camera-Static Filtering. Reliable manipulation supervision requires near-static cameras in each pair. Instead of using optical flow [17] or hand-crafted feature matching [29], we use *camera token clustering*, the camera token z_{cam} in Eq. (5) is predicted by *3D foundation models*. We cluster these tokens with DBSCAN [15] and keep high-density clusters as camera-static clips:

$$C = \text{DBSCAN}\Big(\{z_{\text{cam}}^{(k)}\}_{k=1}^{K}; \epsilon,\ N_{\min}\Big), \tag{16}$$

where ϵ is the neighborhood radius and $N_{\min}$ is the minimum samples per cluster.

Depths & Masks Processing. For each selected cluster, we predict per-frame depth with the 3D foundation model, detect main objects on key frames with an open-world detector [23], and propagate masks through the clip using a video object tracker [8].

Depth-Aware Selection. To select pairs with clear 3D manipulation, we unproject each masked frame $(I^{(k)} * M^{(k)})$ into 3D using depth $D^{(k)}$ and camera pose $(R^{(k)}, t^{(k)})$, then estimate an object representative coordinate with a coordinate-wise median:

$$\mathbf{c}_o^{(k)} = \text{Median}\Big(\text{Unproj}(I^{(k)} * M_o^{(k)}, D^{(k)}, R^{(k)}, t^{(k)})\Big), \tag{17}$$

$$d_o^{(i,j)} = \Big\|\mathbf{c}_o^{(i)} - \mathbf{c}_o^{(j)}\Big\|_2. \tag{18}$$

We choose the frame pair (i, j) with the largest distance $d_o^{(i,j)}$. For short clips, we use the first and last frames to reduce overhead. Finally, we apply depth-threshold filtering and VLM verification [40] to remove samples with incorrect masks or insufficient 3D displacement. Each pair is further annotated with expanded prompts that describe non-geometric appearance changes (e.g., color or background variation), facilitating better text-image alignment for subsequent training.

4 Experiments

4.1 Training Details

We use Qwen-Image-Edit [38] as the DiT editing backbone and Depth-Anything-3 [26] as the 3D foundation model. We fine-tune the DiT with LoRA [18] (rank $r = 128$), mainly on self-attention layers. Training is conducted on 4× NVIDIA RTX PRO 6000 GPUs, with total batch size 64 for 10K steps. We use AdamW [28] with learning rate 1×10^{-4} and weight decay 0.01. The depth-loss weight is set to $\lambda_d = 0.1$.

4.2 Benchmark and Metrics

Data. Since there is no widely accepted benchmark for this task, we build an evaluation set with 200 image pairs and about 320 individual objects. It covers diverse scenes, object categories, and object scales. Half of the pairs contain a single manipulated object, and the other half contain multiple manipulated objects. Each pair includes depth annotations and object-level labels.

Metrics. We report metrics from five aspects.

1. *2D Spatial Accuracy.*

- **DIoU** [61]: Distance IoU between predicted and ground-truth boxes.
- **Mask IoU**: IoU between predicted and ground-truth masks.

2. *Depth Accuracy.* Given predicted depth D^{pred}, ground-truth depth D^{gt}, and valid object pixels Ω:

Table 1: Quantitative comparison on ManipEval. All metrics are linearly normalized to $[0, 100]$. Methods are sorted by Chamfer distance in descending order. Proprietary commercial models are marked with †.

Method	DIoU↑	Mask IoU↑	AbsRel↓	$\delta_{1.25}$↑	Chamfer↓	Centroid↓	RA-DINO↑	DeQA↑	Phys-VLM↑
GeoDiffuser	51.72	20.10	67.64	31.40	73.41	64.56	14.78	56.72	54.62
DiffusionHandles	56.22	18.88	57.36	31.73	73.00	59.65	17.08	61.83	46.90
Move-and-Act	46.89	10.97	68.47	29.30	64.59	63.62	13.01	60.79	60.85
OBJect 3DIT	46.63	10.18	66.93	32.75	62.30	64.14	14.31	48.01	66.65
GoodDrag	51.07	13.29	69.60	34.19	51.27	58.39	20.85	74.08	85.08
PixelMan	49.48	11.49	66.46	35.10	50.18	57.13	21.17	72.79	73.60
Qwen-Image-Edit	53.28	13.80	64.63	40.61	46.31	52.48	26.09	75.67	90.55
LightningDrag	53.81	18.90	57.07	35.99	45.76	53.57	21.92	73.60	88.60
ChronoEdit	48.92	9.26	65.80	36.35	45.02	54.40	21.73	75.29	92.15
GPT-Image-1.5†	52.33	11.34	70.95	36.50	39.79	52.78	23.80	75.82	88.68
Qwen-Image-2.0-Pro†	56.48	15.60	55.93	41.39	29.64	43.48	31.04	68.26	82.91
Nano Banana Pro†	59.97	18.93	55.02	46.11	25.33	35.62	34.77	**77.48**	91.06
Ours	**65.33**	**27.20**	**49.53**	**51.08**	**18.93**	**32.12**	**36.91**	75.48	**93.72**

- **AbsRel** [14]:

$$\mathrm{AbsRel} = \frac{1}{|\Omega|} \sum_{i \in \Omega} \frac{|D_i^{\mathrm{pred}} - D_i^{\mathrm{gt}}|}{D_i^{\mathrm{gt}}}. \tag{19}$$

- $\boldsymbol{\delta}_{1.25}$ [14]: ratio of pixels satisfying $\max\left(\frac{D_i^{\mathrm{pred}}}{D_i^{\mathrm{gt}}}, \frac{D_i^{\mathrm{gt}}}{D_i^{\mathrm{pred}}}\right) < 1.25$.

3. 3D Manipulation Accuracy. We reconstruct object point clouds and compare with ground truth:

- **Chamfer Distance** [2]:

$$\mathrm{CD}(P^{\mathrm{pred}}, P^{\mathrm{gt}}) = \frac{1}{|P^{\mathrm{pred}}|} \sum_{p \in P^{\mathrm{pred}}} \min_{q \in P^{\mathrm{gt}}} \|p - q\|_2 + \frac{1}{|P^{\mathrm{gt}}|} \sum_{q \in P^{\mathrm{gt}}} \min_{p \in P^{\mathrm{pred}}} \|q - p\|_2. \tag{20}$$

  We normalize point clouds by the valid-scene diagonal to make scores comparable.

- **Centroid Distance**:

$$\mathbf{c}^{\mathrm{pred}} = \frac{1}{|P^{\mathrm{pred}}|} \sum_{p \in P^{\mathrm{pred}}} p, \qquad \mathbf{c}^{\mathrm{gt}} = \frac{1}{|P^{\mathrm{gt}}|} \sum_{q \in P^{\mathrm{gt}}} q, \tag{21}$$

$$\mathrm{CentroidDist}(P^{\mathrm{pred}}, P^{\mathrm{gt}}) = \|\mathbf{c}^{\mathrm{pred}} - \mathbf{c}^{\mathrm{gt}}\|_2. \tag{22}$$

4. Image Quality and Consistency.

- **Relocation-Aware DINO Similarity**: DINO similarity [45] penalized by relocation-vector errors. Let $\mathbf{v}^{\mathrm{pred}}$ be decomposed into components parallel and orthogonal to $\mathbf{v}^{\mathrm{gt}}$:

$$e_{\parallel} = \frac{\|\mathbf{v}_{\parallel} - \mathbf{v}^{\mathrm{gt}}\|_2}{\|\mathbf{v}^{\mathrm{gt}}\|_2 + \varepsilon}, \qquad e_{\perp} = \frac{\|\mathbf{v}_{\perp}\|_2}{\|\mathbf{v}^{\mathrm{gt}}\|_2 + \varepsilon}, \tag{23}$$

$$S_{\text{RA-DINO}} = S_{\text{DINO}} \cdot \exp\left(-\alpha e_{\parallel} - \beta e_{\perp}\right). \tag{24}$$

  We set $\alpha = 1$ and $\beta = 0.8$.

- **DeQA Score** [56]: general perceptual quality metric.

5. VLM Physical Plausibility.

- **Phys-VLM**: VLM-based assessment of physical realism and global scene consistency (e.g., lighting/shadows, depth ordering, contacts/occlusions).

4.3 Baselines

We compare with five groups of baselines:
(1) **Drag-based**: *GoodDrag* [59], *LightningDrag* [43];
(2) **Explicit manipulation**: *Move-and-Act* [22], *PixelMan* [21];
(3) **3D-aware editing**: *OBJect 3DIT* [30], *GeoDiffuser* [42], *DiffusionHandles* [35];
(4) **Video-prior**: *ChronoEdit* [54];
(5) **Commercial models**: *Nano Banana Pro* [11], *GPT-Image-1.5* [33], *Qwen-Image-2.0-Pro* [39]. For commercial systems, we use the strongest publicly available reasoning version.

Besides the methods supporting explicit 3D manipulation, we adjust the input for the other methods to perform a more fair comparison. Setting details are shown in the appendix.

4.4 Comparison

Quantitative results on ManipEval are shown in Tab. 1. A consistent pattern is that existing methods with explicit 2D/3D control still lag behind strong commercial models on final manipulation accuracy, with notably larger gaps on depth and 3D geometry than on 2D overlap. Our method substantially narrows this gap and achieves the **best overall performance** across manipulation-related metrics, while also obtaining the highest RA-DINO score and the strongest Phys-VLM among all baselines, including closed-source commercial systems—indicating better physical plausibility and scene-level consistency after the move, beyond what raw layout or depth scores alone capture.

Comparison with Commercial Baselines. Against the strong commercial baseline, Nano Banana Pro, our method improves DIoU by **5.36** points (65.33 vs. 59.97), reduces Chamfer distance by **6.40** points (18.93 vs. 25.33), and improves RA-DINO by **2.14** points

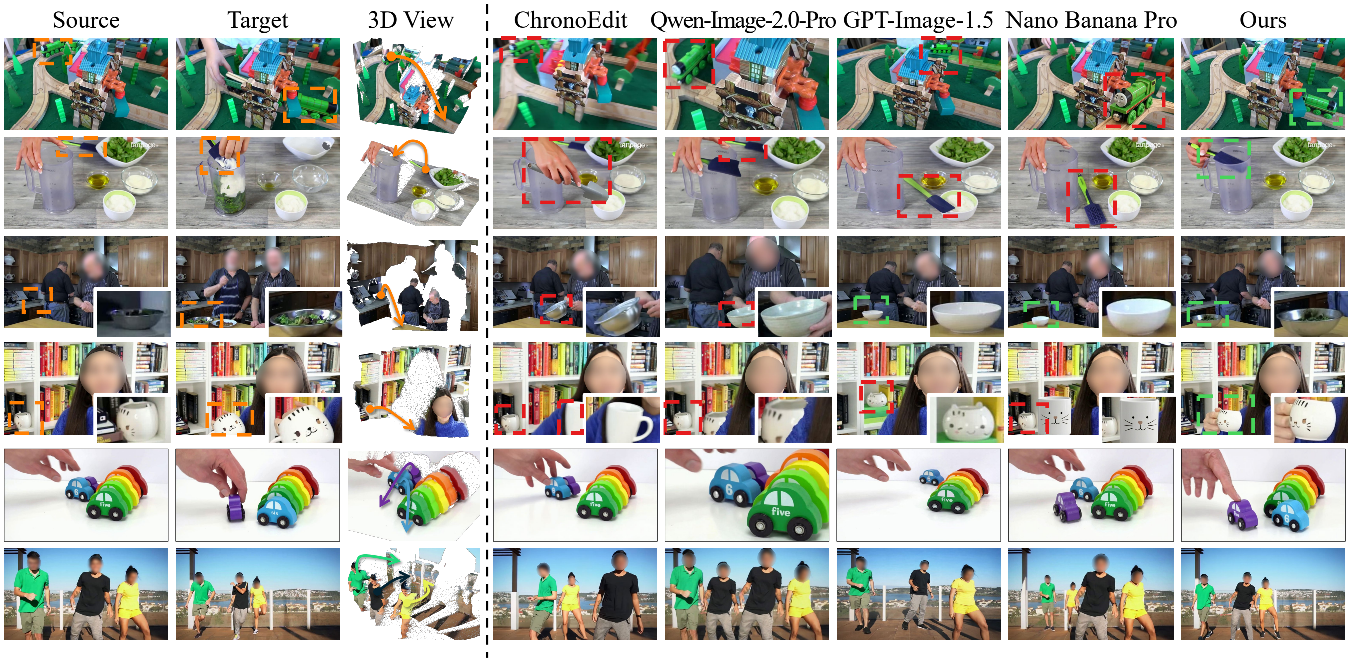

Figure 5: Qualitative comparison on ManipEval. The top two rows highlight manipulation accuracy and geometric consistency; the middle two rows focus on preserving object identity and fine details; the last two rows show multi-object manipulation. In the first four rows, source and target positions are annotated with orange boxes. Objects that are misplaced or still left at the source are marked with red boxes, and correctly manipulated objects are indicated by green boxes.

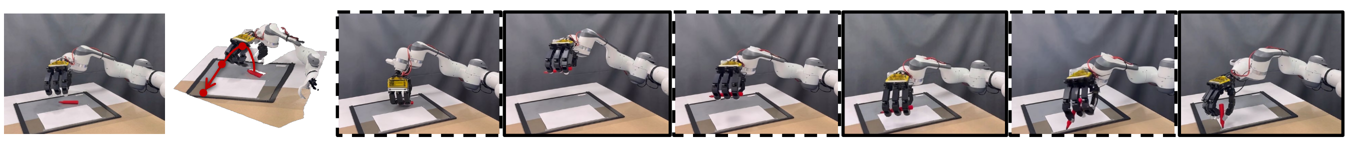

Figure 6: Continuous object manipulation along a trajectory. Given the initial image and a trajectory, PhyEdit performs continuous editing (solid borders). Intermediate frames are further interpolated by a video model (dashed borders). Notably, this robotic arm is out of our training distribution, demonstrating our strong generalization.

(36.91 vs. 34.77). Similar trends are also observed against other commercial systems, especially on geometry-sensitive metrics.

Multi-Object Manipulation. As detailed in the appendix, our lead over Nano Banana Pro grows from 5.19 to 6.58 in Chamfer distance and from 1.91 to 8.02 in $\delta_{1.25}$ from single- to multi-object scenarios. This demonstrates our superior capability to maintain precise 3D geometry in multi-object scenes.

Image Quality Discussion. The DeQA score is close to our base model Qwen-Image-Edit (75.48 vs. 75.67), suggesting that the gains in controllable manipulation and geometric consistency are achieved without noticeable loss of image quality.

Qualitative results are shown in Fig. 5. The first two rows focus on manipulation accuracy and consistency. The middle two rows focus on preserving object identity and fine details. The last two rows focus on multi-object editing.

Manipulation accuracy and consistency. In the first two rows, our method places the toy train and the spatula at the correct 2D and depth locations. Most baselines either fail to move the object, leave a duplicate at the source location, or place the object at an incorrect depth. Our method also handles occlusions more effectively: the train remains partially occluded by the toy building, and the spatula is correctly inserted into the pitcher.

Object identity and fine details. In the third row, the target metal bowl is small. Several baselines did not manipulate the correct object or change its color or texture. In the fourth row, the target mug is also small and partially occluded, and baseline methods struggle to preserve the kitty pattern.

Multi-object editing. In the last two rows, baseline methods often fail to edit all requested objects. Even the strongest commercial baseline, *Nano Banana Pro*, handles only part of the targets, while our method edits all objects consistently.

Continuous Object Manipulation. Beyond single-step editing, PhyEdit is capable of maintaining geometric consistency under continuous state transitions. As illustrated in Fig. 6, given the initial state and spatial trajectory (and optional instructions), our model sequentially renders physically accurate keyframes at specified target

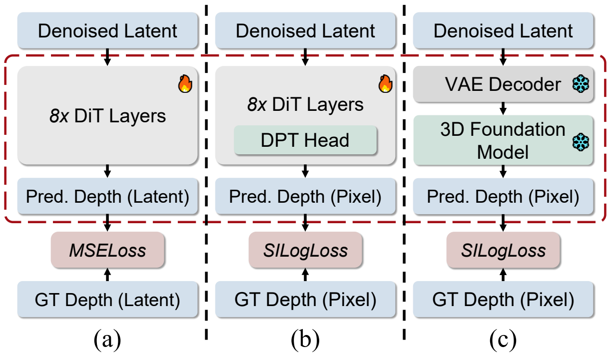

Figure 7: Architectures of different depth supervision methods. The core difference is annotated with red dashed box.

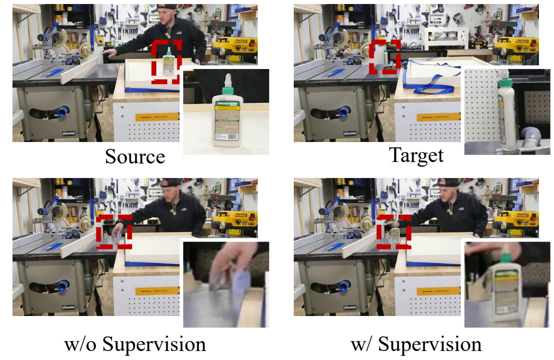

Figure 8: Qualitative comparison of methods with and without depth supervision.

Table 2: Quantitative comparison of different depth supervision methods.

Method	DIoU↑	AbsRel↓	Chamfer↓	RA-DINO↑
w/o Supervision	62.37	50.92	24.52	33.53
Latent-to-Latent	60.48	49.78	28.61	32.29
Latent-to-Depth	<u>64.19</u>	<u>49.73</u>	<u>20.87</u>	<u>35.93</u>
Ours (pixel-level)	**65.33**	**49.53**	**18.93**	**36.91**

Table 3: Quantitative comparison of the effects of reference image and reference-aware training.

Reference	Training	DIoU↑	AbsRel↓	Chamfer↓	RA-DINO↑
✗	✗	45.58	68.22	55.38	18.85
✗	✓	46.16	67.99	52.93	18.87
✓	✗	<u>53.28</u>	<u>64.63</u>	<u>46.31</u>	<u>26.09</u>
✓	✓	**65.33**	**49.53**	**18.93**	**36.91**

points. These sparse frames can then be processed by interpolation models [48] to synthesize a continuous, physically consistent object manipulation video.

4.5 Ablation Studies

Depth Supervision Methods. We compare three depth supervision variants, illustrated in Fig. 7.

(a) Latent-to-Latent Supervision: We train a latent-to-latent module with 8 DiT layers (similar parameter count to DA3 [26]) to predict the latent of the depth map from the image latent. **(b) Latent-to-Depth Supervision**: We add a DPT head [41] on top of the latent-to-latent module in **(a)**. It takes hidden features from several DiT layers and predicts the depth map without decoding the edited image to pixels. **(c) Pixel-level Supervision**: We decode the denoised latent and supervise the depth map of the decoded image (pixel-level alignment with the 3D prior).

Quantitative results are shown in Tab. 2. All depth-supervised variants improve AbsRel over the model without supervision. The gains are around 1 point. However, differences become larger on 2D and 3D metrics. We observe: **1. The Latent-to-Latent variant** is weakest. It drops DIoU by 1.89 points relative to the no-supervision model, and also gives worse Chamfer and RA-DINO. This indicates that pure latent-level depth signals are not precise enough for accurate manipulation. **2. The Latent-to-Depth** variant is stronger and improves all metrics over no supervision, but it is still below our **pixel-level** method (e.g., DIoU 64.19 vs. 65.33, Chamfer 20.87 vs. 18.93). Overall, the pixel-level strategy gives the best balance across 2D, depth, and 3D metrics.

Qualitative results are shown in Fig. 8. In the second row, we compare no depth supervision and pixel-level depth supervision under the same random seed. The target object is very small (smaller than one patch). Without depth supervision, the model only leaves a blurry trace near the target location. With pixel-level depth supervision, the object is generated correctly. This example shows that pixel-level depth cues help the model localize and synthesize small objects more reliably.

Ablation on the Reference Image. As described in Sec. 3.2, we feed a 3D-transformed reference image to the DiT backbone as a visual preview. We ablate both the reference image and the reference-aware training as shown in Tab. 3.

We observe that the reference image is critical. With or without training, using a reference image consistently gives better results than removing it. The benefit of training is also much larger when the reference image is present. Without the reference image, training only reduces Chamfer from 55.38 to 52.93 (-2.45). With reference image, training reduces Chamfer from 46.31 to 24.91 (-21.40).

This ablation reflects the challenge of using text-only control for precise 3D manipulation in current open-source frameworks. While upgrading text encoders can improve spatial instruction understanding—as evidenced by the performance gap between Qwen-Image-2.0-Pro [39] and Qwen-Image-Edit [38]—fine-grained 3D intent remains difficult to accurately recover from language alone. In our setting, providing a reference image supplies explicit spatial context that text cannot reliably convey, leading to superior results. This text-to-spatial bottleneck is where open models still fall behind leading closed-source systems. Narrowing this gap is crucial for achieving better physical plausibility in complex scene manipulation.

5 Conclusion

In this paper, we develop a DiT-based image editing framework, PhyEdit, for 3D-aware object manipulation. The framework combines contextual 3D-aware visual guidance with a joint 2D-3D supervision to improve physical consistency and spatial accuracy. To support this method and evaluate performance, we present a real-world dataset, RealManip-10K, with detailed object-level annotations, and build a benchmark, ManipEval, to evaluate the physical accuracy of object manipulation across multiple dimensions. Experiments show that PhyEdit outperforms existing methods, including strong commercial baselines, and ablation studies validate the contribution of each key component. We believe that achieving physically-grounded image editing is a step towards the broader goal of building interactive world models.

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A Experimental Details

A.1 Training Details

This section adds training details that were omitted from the main paper for space.

Base model and fine-tuning. We take Qwen-Image-Edit-2511 [38] as the backbone; it is a standard choice for image-editing fine-tuning. We add LoRA [18] with rank 128 mainly to the self-attention blocks of the DiT, and use a smaller rank on the remaining layers (e.g., MLPs).

Prompt template. The prompt follows this template:

"Assume the image is in a 3D space where the origin is at the top-left-near corner of the image. The X-axis (left-right) and Y-axis (down-up) range from 0 to 1, while the Z-axis (depth) ranges from 0 to 1 (near to far). Move the <OBJECT_NAME> at <SOURCE_POSITION> to <TARGET_POSITION>, and <OBJECT_EDIT_INSTRUCTIONS>. <ADDITIONAL_INSTRUCTIONS>."

Our pairs come from real videos, so edits are not limited to the object positions. We use an LLM [40] to compare the source and target images and generate the <OBJECT_EDIT_INSTRUCTIONS> and <ADDITIONAL_INSTRUCTIONS> for changes beyond pure relocation.

Depth estimation. Depth maps are produced with Depth-Anything-3 [26]. Single-view depth estimation is not reliable enough for our setting, so at training time we always feed the model an image pair from the same edit. The edited image is paired with the source to obtain $\mathbf{D}_{\text{edit}}$; the target is paired with the source to refresh the ground-truth depth $\mathbf{D}_{\text{gt}}$ for the source view. In symbols,

$$\mathbf{D}_{\text{src}}, \mathbf{D}_{\text{edit}} = \text{DA3}([\mathbf{I}_{\text{src}}, \mathbf{I}_{\text{edit}}]), \tag{25}$$

$$\mathbf{D}_{\text{src}}, \mathbf{D}_{\text{gt}} = \text{DA3}([\mathbf{I}_{\text{src}}, \mathbf{I}_{\text{tgt}}]), \tag{26}$$

where $\mathbf{D}_{\text{edit}}$, $\mathbf{D}_{\text{src}}$, and $\mathbf{D}_{\text{gt}}$ are depth for the edited frame, the shared source estimate, and the target-aligned ground truth, and $\mathbf{I}_{\text{src}}$, $\mathbf{I}_{\text{edit}}$, $\mathbf{I}_{\text{tgt}}$ are the corresponding RGB inputs. Depth supervision uses $\mathbf{D}_{\text{edit}}$ against $\mathbf{D}_{\text{gt}}$.

Depth is predicted at roughly the source resolution. We use the same paired-input procedure at evaluation time.

A.2 Experimental Settings for Baselines

We group baselines by how spatial control is given:

(1) Explicit 2D manipulation. These methods expect explicit 2D cues. We supply masks, coordinates, and bounding boxes as required. This set includes *GoodDrag* [59], *LightningDrag* [43], *OBJect 3DIT* [30], *Move-and-Act* [22], and *PixelMan* [21].

(2) Explicit 3D manipulation. These methods take 3D-style controls (e.g., 3D coordinates or camera pose). We provide the 3D inputs each implementation needs. This set includes *GeoDiffuser* [42] and *DiffusionHandles* [35].

(3) Implicit editing. These systems do not accept explicit spatial handles. We spell out 3D spatial information in text using the template in Sec. A.1, and we add source images marked with the object boxes and overlays for the target regions to guide the model to understand the editing context and perform the editing. This set includes *ChronoEdit* [54], *GPT-Image-1.5* [33], *Qwen-Image-2.0-Pro* [39], and *Nano Banana Pro* [11].

Most explicit-control baselines do not accept an extra free-text prompt beyond spatial inputs. So we omit any additional instructions for all baselines. For a baseline that only supports manipulating one object at a time, we run it iteratively until all objects are handled. Overall, we tailor inputs for each baseline and tune its interface so it receives as complete spatial information as its API allows.

A.3 Details on Metrics

Localizing manipulated objects. Locating the moved object in generated images is challenging. Feature-based matching [44] is too coarse for grounding, and generic detectors [8] can be unreliable on edited content. Some baselines may leave a copy at the source, which can confuse the detectors and lead to suboptimal object localization. We use a two-stage procedure: Qwen-3.5-Plus [40] proposes coarse boxes with multimodal reasoning, and SAM3 [8] refines them to masks.

We pass the ground-truth boxes from the benchmark dataset into the prompt and ask Qwen-3.5-Plus to look for the manipulated object near the target region first, then outside it, and to return a short phrase that uniquely describes that object. SAM3 then refines the coarse box with that phrase to produce the mask.

Penalizing missing objects. If the VLM cannot find the object or the mask is empty—sometimes happens when the edit removes the object or collapses to severe artifacts—we still count the sample instead of dropping it. Accuracy-style scores are set to zero for that case. Distance-style scores (Chamfer, centroid, etc.) use a data-driven penalty derived from the global distribution of successful localization:

$$p = \max(q_{0.99}, 1.2 \cdot q_{0.95}), \tag{27}$$

where q_n is the n-th percentile of the empirical distribution over the other manipulated objects which are successfully localized.

A.4 More Qualitative Results on ManipEval

Additional qualitative comparisons are shown in Fig. 9.

Manipulation accuracy and consistency. In the first two rows most baselines miss the correct placement. *GPT-Image-1.5* (row 1) and *Qwen-Image-2.0-Pro* (row 2) sometimes move the object but mainly rescale it. The result of the manipulation looks like a 2D scale change rather than a true depth shift.

Object identity and fine details. In rows 3–6 depth placement is often wrong. In rows 4–5 *Nano Banana Pro* is closer, yet lighting or scale can look slightly unrealistic.

Multi-object editing. In the last two rows many baselines fail to handle part of the manipulation requests. Even *Nano Banana Pro* finishes only a subset of objects, whereas our method moves each target as requested consistently.

A.5 Results on ManipEval Split by Object Number

Multi-object scenes in ManipEval contain two to four edited objects. Compared to the single-object split, depth changes are not large for every object in a scene; however, at least one object has a clear depth shift, and the average depth change remains quite substantial.

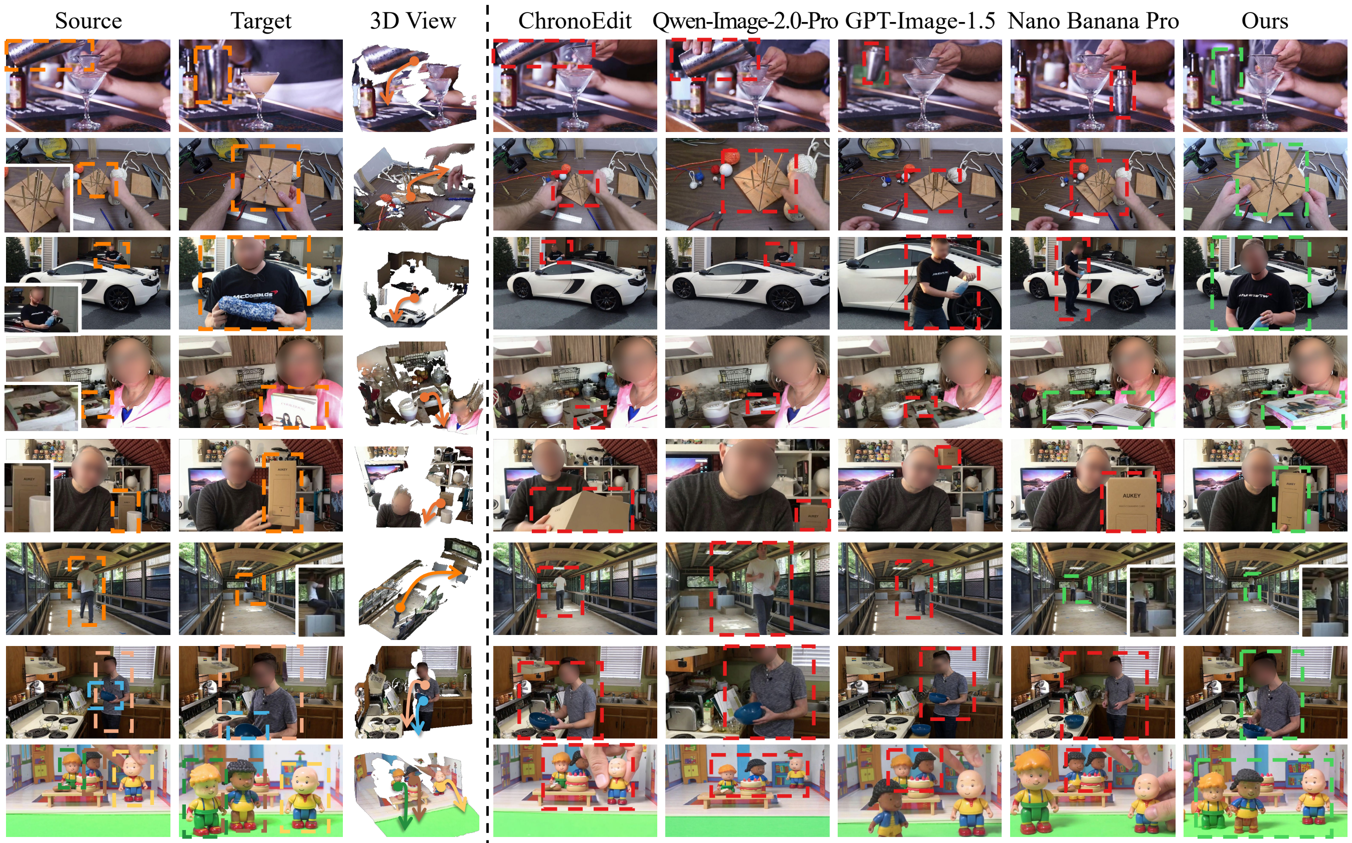

Figure 9: More qualitative results on ManipEval. For baseline methods, misplaced objects and objects still left at the source are marked with red boxes, and correctly manipulated objects are indicated by green boxes.

Table 4: Quantitative comparison on ManipEval on single-object items, sorted by Chamfer distance in descending order.

Method	DIoU↑	Mask IoU↑	AbsRel↓	$\delta_{1.25}$↑	Chamfer↓	Centroid↓	RA-DINO↑	DeQA↑	Phys-VLM↑
GeoDiffuser	51.36	18.70	85.67	24.72	93.34	73.95	15.40	60.23	66.08
Object 3DIT	45.34	10.03	87.47	26.78	91.11	77.96	14.64	48.70	69.70
Move-and-Act	50.54	12.65	83.88	26.76	73.83	68.99	15.32	64.20	66.70
LightningDrag	51.99	17.94	74.68	27.84	66.65	64.31	21.93	72.80	92.00
Qwen-Image-Edit	51.41	12.70	86.77	36.54	66.43	64.38	25.56	74.76	92.80
GoodDrag	51.05	13.65	88.85	31.55	65.55	67.10	22.69	74.20	91.50
DiffusionHandles	58.73	20.21	67.91	27.72	65.54	58.20	20.80	64.25	52.70
PixelMan	50.10	11.81	84.13	34.65	62.98	66.13	22.53	72.61	83.30
ChronoEdit	48.79	9.28	85.58	32.55	58.09	62.37	23.82	73.87	91.85
GPT-Image-1.5†	51.57	11.55	91.56	33.31	50.82	60.91	23.62	74.02	87.24
Qwen-Image-2.0-Pro†	56.34	15.63	69.77	36.20	39.11	49.68	30.98	67.11	81.21
Nano Banana Pro†	63.35	22.40	69.16	44.20	26.45	34.91	38.49	**76.37**	90.20
Ours	**66.29**	**29.10**	**66.66**	**46.11**	**21.26**	**34.45**	**39.14**	74.37	**93.23**

Quantitative results for each split are in Tabs. 4 and 5. The two splits are not directly comparable, but the tables suggest the gap versus strong baselines can shift with the number of objects.

On the single-object split, our method is ahead on the manipulation-oriented columns in Tab. 4; exact values are in the table. Relative to Nano Banana Pro, Chamfer distance improves by 5.19 points and

Table 5: Quantitative comparison on ManipEval on multi-object items, sorted by Chamfer distance in descending order.

Method	DIoU↑	Mask IoU↑	AbsRel↓	$\delta_{1.25}$↑	Chamfer↓	Centroid↓	RA-DINO↑	DeQA↑	Phys-VLM↑
DiffusionHandles	53.72	17.55	46.81	35.74	79.35	59.70	13.35	59.42	41.10
GeoDiffuser	52.07	21.49	49.79	38.01	61.58	55.14	14.17	53.24	43.27
Move-and-Act	43.25	9.29	53.05	31.85	56.91	56.97	10.69	57.39	55.00
Object 3DIT	47.92	10.33	46.39	38.72	40.29	51.06	13.99	47.31	63.60
GoodDrag	51.08	12.92	50.16	36.85	37.65	49.17	18.98	73.96	78.59
PixelMan	48.87	11.17	48.79	35.55	37.62	47.63	19.81	72.96	63.90
ChronoEdit	49.05	9.24	46.02	40.16	32.62	46.39	19.65	76.71	92.45
LightningDrag	55.63	19.86	39.45	44.13	29.81	43.44	21.91	74.41	85.20
Qwen-Image-Edit	55.14	14.91	42.49	44.68	28.97	41.02	26.62	76.58	88.30
GPT-Image-1.5†	53.09	11.14	50.55	39.65	28.87	44.72	23.97	77.61	90.10
Nano Banana Pro†	56.55	15.43	40.74	48.04	23.75	36.13	31.00	**78.61**	91.92
Qwen-Image-2.0-Pro†	56.62	15.57	42.22	46.54	21.68	37.62	31.11	69.39	84.60
Ours	**64.38**	**25.30**	**32.40**	**56.06**	**17.17**	**29.80**	**34.68**	76.58	**94.21**

Absolute Relative Error decreases by 2.50 points, with consistent gains also on DIoU, Mask IoU and Phys-VLM.

On the multi-object split (Tab. 5), the ordering changes, and the same comparison against Nano Banana Pro shows slightly larger differences: Chamfer from 5.19 to 6.58, and $\delta_{1.25}$ from 1.91 to 8.02. This is consistent with the fact that multi-object scene edits require more precise depth and placement consistency. In this setting, our method maintains better physical accuracy and a clearer advantage.

Leading closed-source models, especially Nano Banana Pro, can generate high-quality images. But as shown in the qualitative examples above, its results can still have unrealistic lighting or object scale after manipulation, which hurts physical realism. Phys-VLM shows a similar trend across the two splits.

Overall, this split-by-object-number analysis shows that our method keeps an advantage on geometry-sensitive metrics in both settings, and that this advantage is more pronounced in the multi-object case.

B Limitations and Future Work

Although our method performs well on the benchmark, several limitations remain.

Insufficient prompt control. As discussed in Sec. A.1, changes in real-video image pairs are not always fully aligned with the additional prompt and the text may not cover all visible edits. In addition, our current training does not include dedicated optimization for the additional free-form text branch. As a result, prompt control from free-text additional instructions is still not that strong, especially for fine-grained constraints about where to edit and where to preserve content.

Failure to handle extreme cases. Our method can still fail in difficult scenes. Artifacts are more likely when the request is highly complex, the scene is heavily cluttered, or the target motion is geometrically extreme (e.g., moving an object too close to the camera, where parts may fall behind the lens, or where projected geometry becomes too dense). Likewise, when an object is moved from far distance to near distance, appearance details may not be recovered reliably.

To address these issues, we plan to improve both controllability and robustness in future work.

Enhance text-visual co-prompting. Our current model relies more on transformed visual guidance than on text-only control. The ablation in the main paper shows that text-only fine-tuning is weaker than joint text+visual conditioning, likely due to limited text-side capacity in the current text encoder. In future work, we will strengthen text-visual co-prompting by improving the text encoder and injecting text conditions more directly into visual guidance. We will also explore methods that infer 3D transformations directly from text prompts, so that visual guidance can be synthesized automatically from language instructions.

Enhance manipulation robustness. To improve robustness, we will scale the training data with more extreme motions and more complex scenes. We also plan to explore reinforcement-learning-based tuning to better align edited results with user intent and to improve physical consistency under hard cases.

Expand to more tasks. We also plan to extend the framework to broader downstream settings. Examples include trajectory-conditioned video manipulation, tighter integration with 3D generation/manipulation pipelines, and faster inference so the model can serve as a key-state renderer in world model systems.

Overall, these directions aim to make the method more controllable, more robust in challenging geometry, and more practical for real applications.